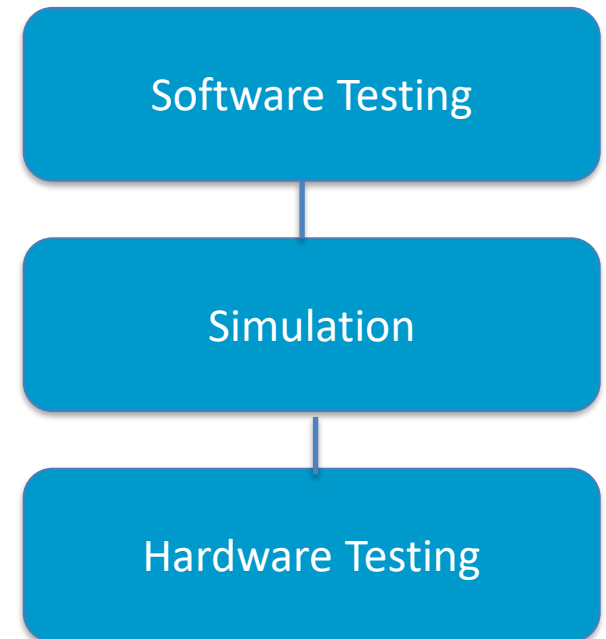


Debugging Techniques in Embedded Systems

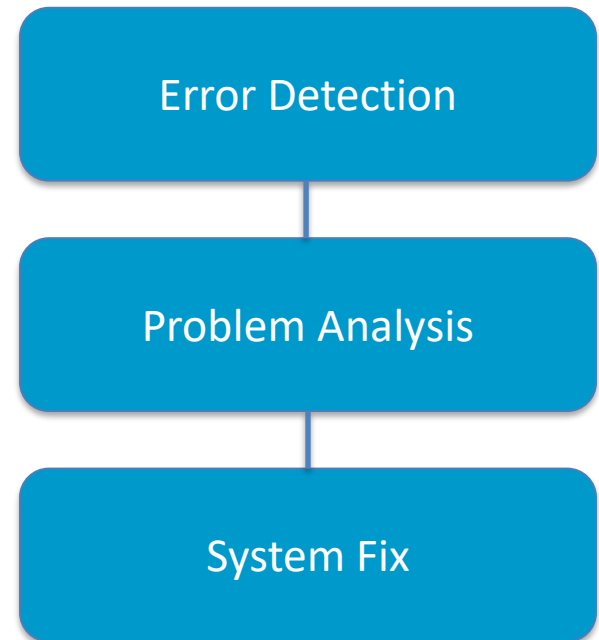
Debugging Techniques

- Debugging means finding and fixing errors in hardware or software.
- In embedded systems hardware and software work together.
- If a problem occurs it can come from either side.
- So developers test each part separately before combining them.
- Debugging helps developers trace the root cause of errors by analyzing program execution step by step.
- It also enables engineers to test system behaviour under different operating conditions before final deployment.



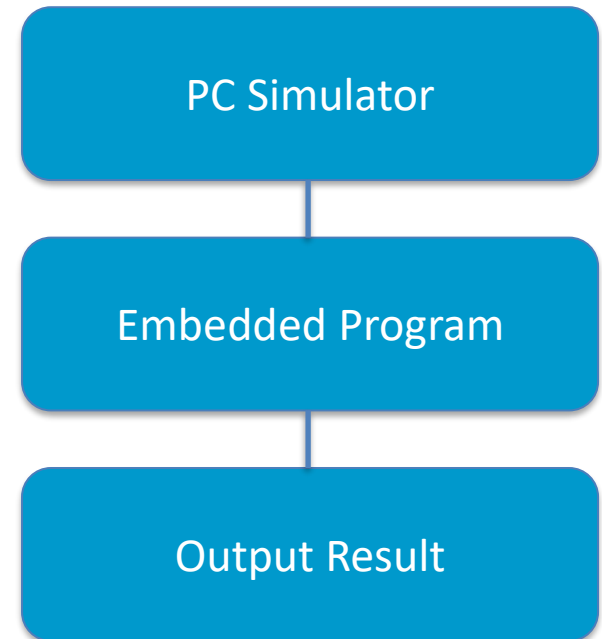
Importance of Debugging

- Debugging helps developers detect errors early.
- It finds logic errors, timing problems and memory faults.
- It prevents system failure in the final product.
- It improves reliability and performance of embedded systems.
- Debugging helps developers identify hidden system faults that may appear during real-time operation.
- It ensures that the embedded system meets design specifications and functions correctly in all conditions.



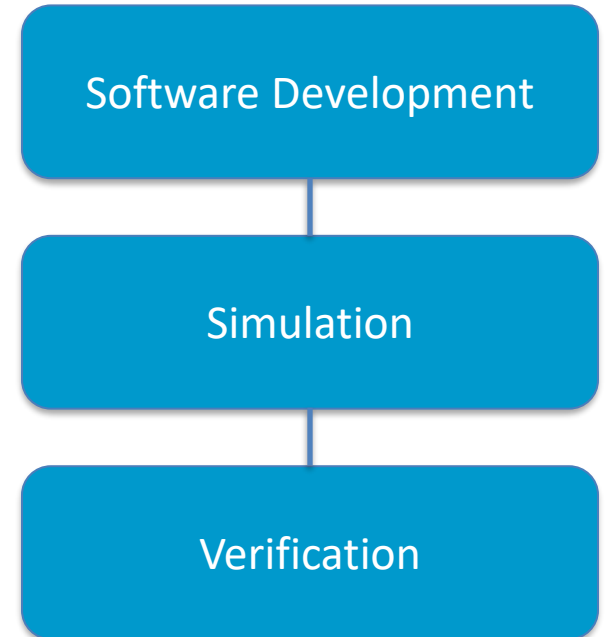
High Level Language Simulation

- This technique tests embedded programs on a computer.
- Programs are written in C or C++ and run on a PC.
- Inputs and outputs are simulated using software.
- This helps test program logic before hardware is ready.
- It helps developers detect logical errors in the program before transferring the code to the embedded hardware.
- This method also allows testing different input conditions and verifying the program output easily.



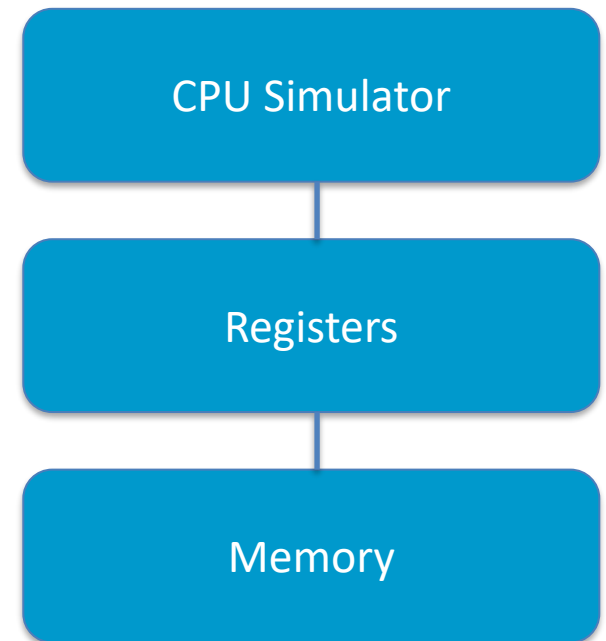
Advantages of High Level Simulation

- Software can be developed even before hardware is built.
- It allows fast program testing and modification.
- It reduces development time and cost.
- It helps verify program logic easily.
- It allows developers to analyze and debug complex algorithms easily using powerful computer tools.
- It also helps reduce development risk by identifying software errors before hardware implementation.



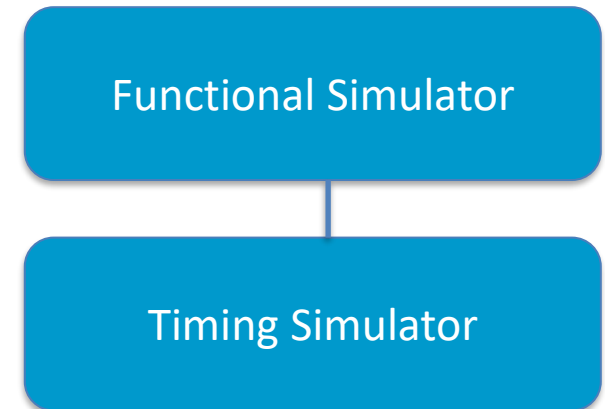
Low Level Simulation

- Low level simulation focuses on processor instructions.
- It allows testing of assembly language programs.
- It shows register operations and memory access.
- This helps understand how the processor executes code.
- It allows developers to analyze the interaction between processor instructions and hardware components.
- This method helps engineers verify low-level routines such as device drivers and interrupt handling programs.



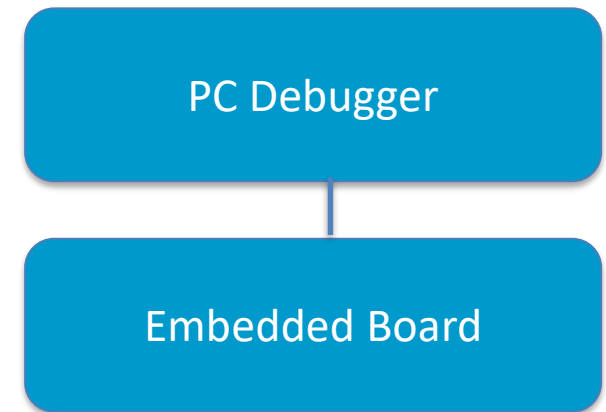
Types of Low Level Simulators

- Functional simulator: Simulates processor operations.
- It allows developers to step through instructions.
- Timing simulator: Shows clock cycles and timing behavior.
- It helps analyze system performance.
- These simulators help developers observe processor operations without using the actual hardware system.
- They are useful for studying system performance and verifying the correctness of low-level program execution.



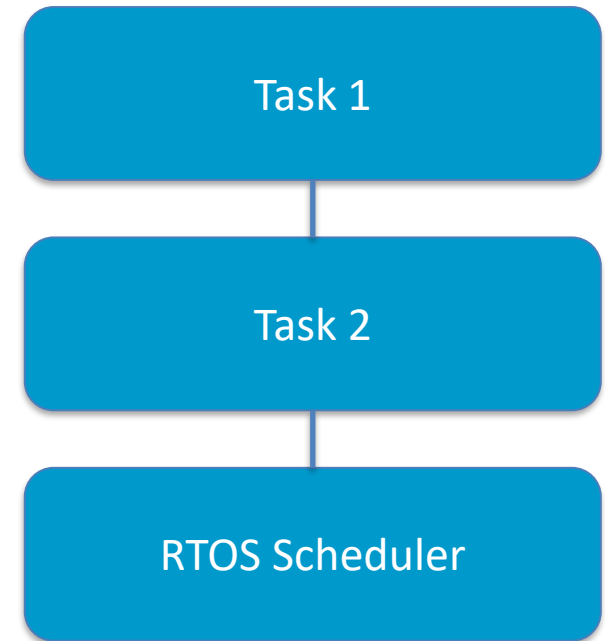
Onboard Debugger

- An onboard debugger runs directly on the embedded board.
- It initializes hardware and controls program execution.
- It supports breakpoints and memory inspection.
- It also allows downloading code into the board.
- It allows developers to pause program execution and examine system behaviour during runtime.
- Onboard debuggers are commonly used in microcontroller development boards for testing embedded firmware.



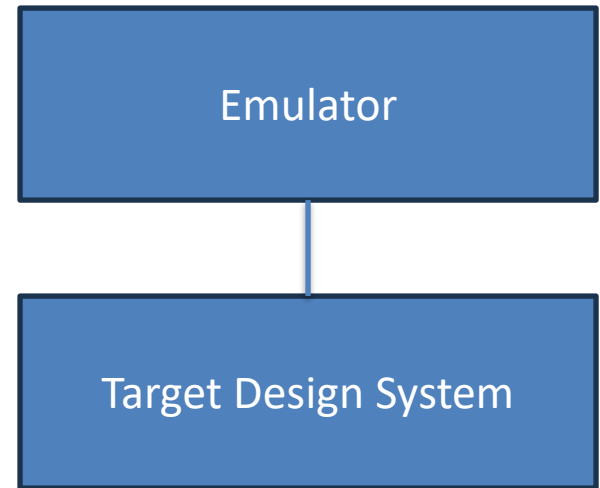
Task Level Debugging

- This debugging is used when an RTOS is present.
- It allows developers to debug tasks instead of instructions.
- It monitors task communication and interrupts.
- It helps detect scheduling problems.
- It enables developers to observe task scheduling and execution order in real-time operating systems.
- This technique helps identify deadlocks, priority inversion, and resource-sharing problems between tasks.



Emulation in Design System

- Emulation in design systems allows developers to test hardware behaviour before the final product is built.
- It replicates the processor and peripheral functions using a special emulator.
- Engineers can analyze timing, signals and memory activity in real time.
- This helps detect design errors early in the development process.
- Emulation helps engineers test and validate system design before the final hardware is manufactured.
- It enables designers to detect hardware and software integration issues early in the development process.



Emulation and Xray Debugging

- Emulation uses a hardware device called an emulator.
- It behaves like the real processor of the system.
- Developers can monitor registers, memory and signals.
- Xray debugging integrates multiple debugging tools.
- Emulation allows developers to observe internal processor operations and system signals in real time.
- Xray debugging systems provide a unified interface to control and monitor multiple debugging tools efficiently.

